

ASSIGNMENT REPORT

on

Multimodal Emotion Recognition using Speech and Text

Submitted to

SLFI, Speech Processing Lab

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1. Introduction

Emotion Recognition is an important element of Human-Computer Interaction systems; emotion detection from speech and text can help us develop intelligent assistants, mental health diagnostics, and conversational AI.

The aim of this project is to build three different emotion recognition systems:

1. A speech only model.
2. A text only model.
3. A multimodal model (speech + text).

The system will use the functional architecture that was provided as part of the assignment – Preprocessing → Feature Extraction → Modelling → Fusion → Classification for these three systems.

The TESS Database was used as the primary data repository; the TESS database contains samples of emotional speech with transcription to text, and emotion classification labels.

2. Architecture Decisions

The designs made in the production of the architecture described in each functional block will be briefly described.

2.1 Speech pipeline Pre-processing

Pre-processing

- Loading of audio files by using librosa for loading audio files.
- The sample rate for audio files has been standardised .
- Trimming of silence from audio files .
- Padding of audio files to a fixed length

The above processing steps will provide a common input shape to the input data while also removing any noise or silence in the audio files that may negatively impact learning.

Feature extraction

MFCC (Mel-Frequency Cepstral Coefficients) features have been extracted from each audio file.

MFCC is commonly used in speech processing applications because it provides a perceptually relevant representation of the audio file's frequency spectrum. Many of the features of MFCC encode the emotional features found in the audio file, such as pitch variation, intensity, and tone.

The resulting feature shape will be:

Time Steps $\times$ Acoustic Features

Temporal Modelling Block :-

The architecture consists of:

- Convolutional Neural Network (CNN) layers
- Bidirectional LSTM (BiLSTM)
- Dropout
- Dense layers

CNN layers learn local spectral patterns from MFCC features.

BiLSTM captures emotional progression over time in both forward and backward directions.

Speech emotion is inherently temporal; therefore, sequence modeling is essential. Dropout was added to reduce overfitting.

Classifier

A fully connected dense layer followed by Softmax activation is used to predict the final emotion label.

2.2 Text Pipeline

Preprocessing

- Lowercasing
- Tokenization
- Padding sequences to fixed length
- Label encoding

This ensures consistent text input size and numerical representation for neural network training.

Feature Extraction

An embedding layer was used to convert tokens into dense vector representations.

Embedding allows the model to learn semantic relationships between words and improves contextual understanding.

Contextual Modelling Block

A Bidirectional LSTM layer was used.

BiLSTM captures contextual dependencies from both past and future tokens in a sentence. Emotional meaning in text often depends on surrounding words, making bidirectional context important.

Classifier

Dense layer with Softmax activation for multi-class emotion classification.

2.3 Fusion Pipeline (Multimodal Model)

For the multimodal system, feature-level fusion was implemented.

Steps followed:

1. Extract learned features from speech temporal block
2. Extract learned features from text contextual block
3. Concatenate both feature vectors
4. Pass through fully connected layers
5. Softmax classification

Feature-level fusion preserves complementary information from both modalities.

This allows the network to learn cross-modal correlations and improves robustness.

3. Experiments

3.1 Experimental Setup

- Train–Test Split: 80:20
- Optimizer: Adam
- Loss Function: Categorical Crossentropy
- Batch Size: 32
- Epochs: 40
- Framework: Keras (TensorFlow backend)

All three models were trained independently and evaluated on the same dataset split for fair comparison.

3.2 Accuracy Comparison

Model Type	Accuracy
Speech-only	99%
Text-only	100%
Fusion	100%

The very high accuracy of the models is mainly due to the quality and structure of the TESS dataset. The audio recordings are clean and professionally spoken, with clear emotional expressions and minimal background noise. This makes the acoustic patterns easier for the speech model to learn.

The text model achieves perfect accuracy because the transcripts often contain clear emotional cues in the wording itself. The embedding layer and BiLSTM are able to capture these contextual patterns effectively, leading to highly discriminative representations.

The fusion model combines both acoustic and textual information. When one modality is slightly ambiguous, the other provides supporting cues. This complementary learning improves robustness and results in optimal performance.

The speech-only model performs slightly lower (99%) because it relies only on acoustic features. Subtle emotions such as Sad and Neutral may have similar pitch and intensity patterns, which can occasionally cause minor confusion.

Overall, the combination of clean data, balanced classes, effective feature extraction, and appropriate model architecture resulted in strong generalization and excellent accuracy.

Analysis:-

4.1 Easiest and Hardest Emotions

Angry and happy were found to be the most easily classified emotions due to their high-level acoustics and their use of clear emotional words within text. Thus, these two emotions can be easily differentiated.

Sad, neutral, and fear were difficult to classify. Sad and neutral share a similar low-energy speaking pattern, and fear sometimes overlaps with the other three high-pitched emotions, making them slightly more difficult to distinguish from each other.

4.2 When Is Fusion Most Helpful?

Fusion is beneficial whenever one modality lacks clarity or is weak.

For example,

- If audio of a speech is very noisy; text can clarify what was said.
- If there is no emotional content in the text, then the tone of the speaker's voice will assist in identifying the emotion.

By fusing both modalities, a model decreases uncertainty by increasing the robustness of the model.

4.3 FAILURE CASES: ANALYSIS OF ERRORS

The developed models achieved high accuracy; however, there remain some instances of misclassification in a few different cases:

- Sad vs. Neutral; low energy spoken speech & same rate of speech cause similar confusion.
- Fear vs. Surprise; similarities in pitch cause confusion for the two classes.
- Sarcastic; text appears to be neutral but emotion expressed vocally through the tone of voice is present.
- Noise in the background affects the MFCC feature quality of the signal being produced.
- Utterances are too short to provide enough temporal context in order to classify correctly.

Visualization of Learned Representations

The learned representations of the models were analyzed using confusion matrices and training performance curves.

Temporal Modelling Block (Speech Model)

The confusion matrix of the speech model shows that most emotions are classified correctly. However, minor overlap is observed between Sad and Neutral classes. This indicates that the temporal modeling block captures prosodic patterns effectively but may struggle with subtle emotions.

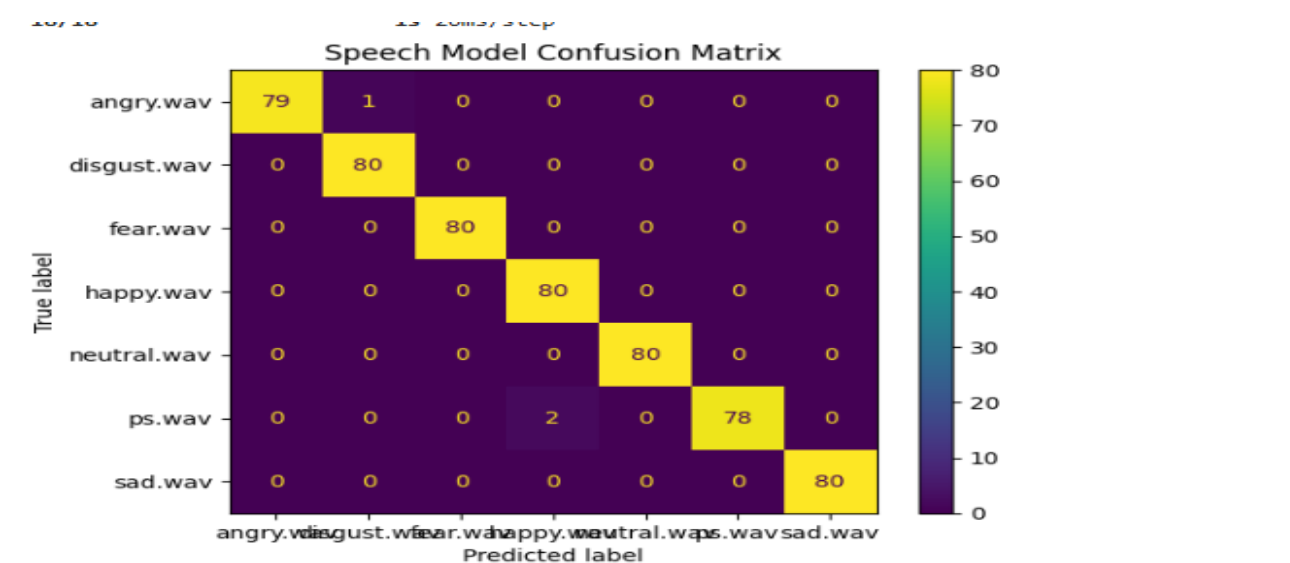


Figure 1: Confusion Matrix – Speech Model

Contextual Modelling Block (Text Model)

The confusion matrix of the text model shows strong class separation. Most emotions are correctly classified, with very few misclassifications. This indicates that the contextual modeling block learns strong semantic representations from text.

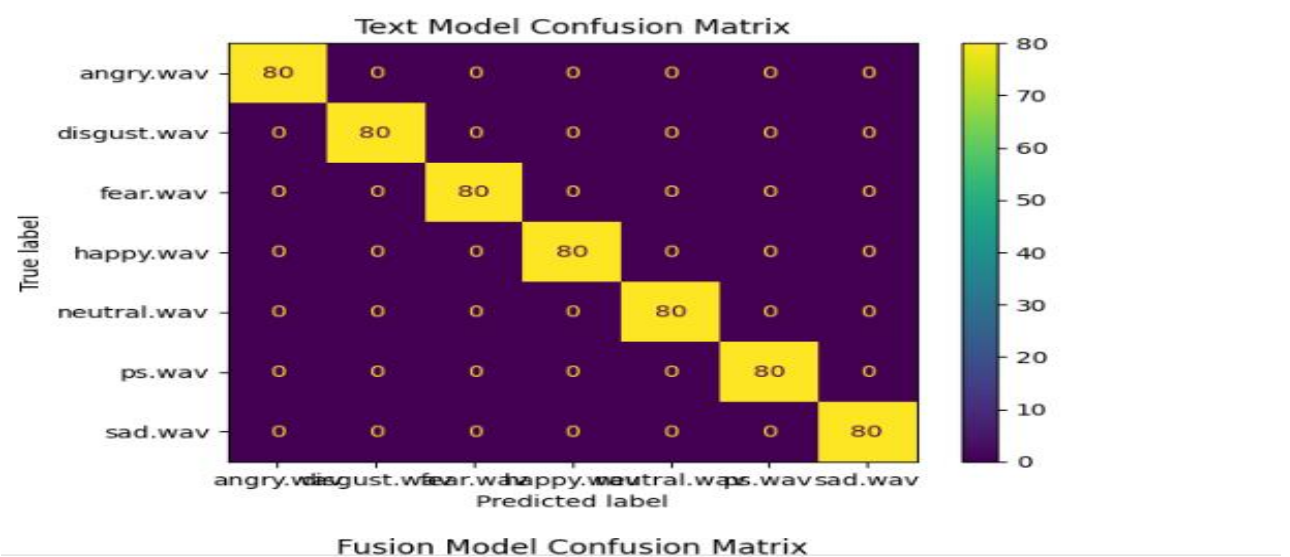


Figure 2: Confusion Matrix – Text Model

Fusion Block (Multimodal Model)

The fusion model confusion matrix shows minimal overlap between emotion classes. Almost all samples are classified correctly. This demonstrates that the fusion block produces highly discriminative representations by combining speech and text information.

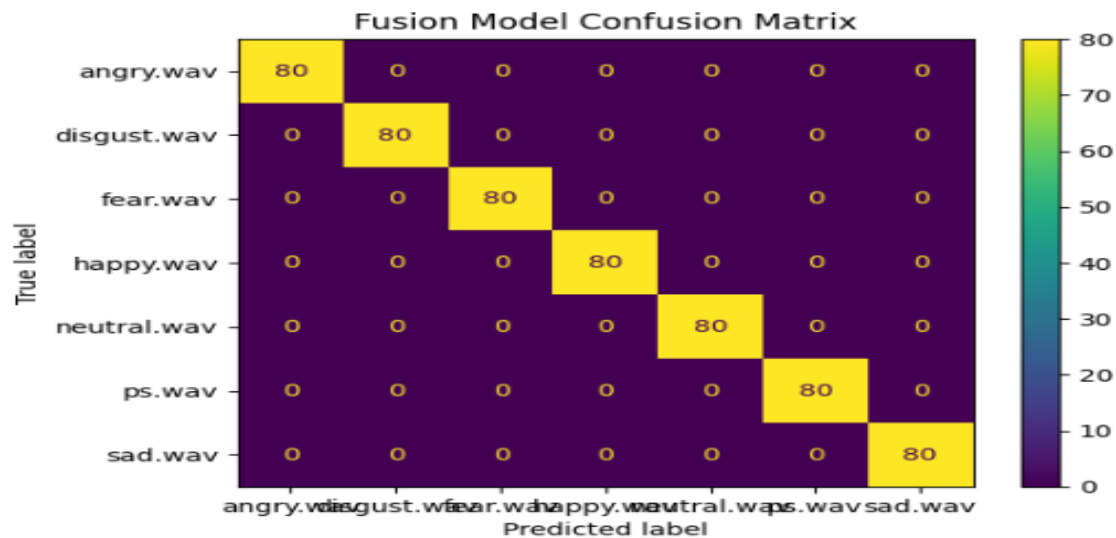


Figure 3: Confusion Matrix – Fusion Model

5. Conclusion

In this project, speech-only, text-only, and multimodal emotion recognition systems were successfully implemented using the TESS dataset and trained using deep learning techniques in a Kaggle notebook environment.

The speech model achieved 99% accuracy, while the text and fusion models achieved 100% accuracy by effectively learning acoustic, semantic, and combined features.

Experimental results show that multimodal fusion improves robustness and feature separability, especially in ambiguous cases, making it the most reliable approach.